

Day 11 - File Ownership Challenge

Topics Covered

- File Ownership
 - Groups
 - chown
 - chgrp
 - Recursive Ownership (-R)
-

Task 1: Understanding File Ownership

Step 1: Check Files

Run:

```
ls -l
```

Output:

```
-rw-r----- 1 ubuntu ubuntu 12 May 31 14:49 notes.txt
```

Format:

```
-rw-r--r-- 1 owner group size date filename
```

```
ubuntu@ip-172-31-44-9:~$ ls -l
total 12
-r--r--r-- 1 ubuntu ubuntu    0 May 31 14:46 devops.txt
-rw-r----- 1 ubuntu ubuntu   12 May 31 14:49 notes.txt
drwxr-xr-x 2 ubuntu ubuntu 4096 May 31 15:06 project
-rw-rw-r-- 1 ubuntu ubuntu   13 May 31 14:52 script.sh
ubuntu@ip-172-31-44-9:~$
```

Difference Between Owner and Group

Owner

The primary user who controls the file.

ubuntu

Group

A collection of users who can share access to the file.

Example:

devops-team

Task 2: Basic chown Operations

Step 1: Create File

```
touch devops-file.txt
```

Step 2: Check Owner

```
ls -l devops-file.txt
```

```
ubuntu@ip-172-31-44-9:~$ ls -l devops.txt
-r--r--r-- 1 ubuntu ubuntu 0 May 31 14:46 devops.txt
```

Step 3: Create Users

Create user tokyo

```
sudo useradd tokyo
```

Create user berlin

```
sudo useradd berlin
```

```
ubuntu:x:1000:1000:Ubuntu:/home/ubuntu:/bin/bash
berlin:x:1001:1001::/home/berlin:/bin/sh
tokyo:x:1002:1002::/home/tokyo:/bin/sh
```

Step 4: Change Owner to tokyo

```
sudo chown tokyo devops-file.txt
```

Verify:

```
ls -l devops-file.txt
```

Example Output:

```
-rw-r--r-- 1 tokyo ubuntu 0 May 27 devops-file.txt
```

```
ubuntu@ip-172-31-44-9:~$ sudo chown tokyo devops.txt
ubuntu@ip-172-31-44-9:~$ ls l
ls: cannot access 'l': No such file or directory
ubuntu@ip-172-31-44-9:~$ ls -l
total 12
-r--r--r-- 1 tokyo  ubuntu    0 May 31 14:46 devops.txt
-rw-r----- 1 ubuntu ubuntu   12 May 31 14:49 notes.txt
drwxr-xr-x 2 ubuntu ubuntu 4096 May 31 15:06 project
-rw-rw-r-- 1 ubuntu ubuntu   13 May 31 14:52 script.sh
ubuntu@ip-172-31-44-9:~$
```

Step 5: Change Owner to berlin

```
sudo chown berlin devops-file.txt
```

Verify:

```
ls -l devops-file.txt
```

```
ubuntu@ip-172-31-44-9:~$ ls -l
total 12
-r--r--r-- 1 berlin ubuntu    0 May 31 14:46 devops.txt
-rw-r----- 1 ubuntu ubuntu  12 May 31 14:49 notes.txt
drwxr-xr-x 2 ubuntu ubuntu 4096 May 31 15:06 project
-rw-rw-r-- 1 ubuntu ubuntu   13 May 31 14:52 script.sh
ubuntu@ip-172-31-44-9:~$
```

Task 3: Basic chgrp Operations

Step 1: Create File

```
touch team-notes.txt
```

Step 2: Check Group

```
ls -l team-notes.txt
```

```
ubuntu@ip-172-31-44-9:~$ ls -l team-notes.txt
-rw-rw-r-- 1 ubuntu ubuntu 0 May 31 17:09 team-notes.txt
ubuntu@ip-172-31-44-9:~$
```

Step 3: Create Group

```
sudo groupadd heist-team
```

Step 4: Change Group

```
sudo chgrp heist-team team-notes.txt
```

Step 5: Verify

```
ls -l team-notes.txt
```

```
ubuntu@ip-172-31-44-9:~$ sudo useradd -m heist-team
ubuntu@ip-172-31-44-9:~$ sudo chgrp heist-team team-notes.txt
ubuntu@ip-172-31-44-9:~$ ls -l team-notes.txt
-rw-rw-r-- 1 ubuntu heist-team 0 May 31 17:09 team-notes.txt
ubuntu@ip-172-31-44-9:~$
```

Example Output:

```
-rw-r--r-- 1 ubuntu heist-team 0 May 27 team-notes.txt
```

Task 4: Combined Owner and Group Change

Step 1: Create File

```
touch project-config.yaml
```

Step 2: Create User

```
sudo useradd professor
```

Step 3: Change Owner and Group Together

```
sudo chown professor:heist-team project-config.yaml
```

Verify:

```
ls -l project-config.yaml
```

```
ubuntu@ip-172-31-44-9:~$ sudo chown professor:heist-team project-config.yml
ubuntu@ip-172-31-44-9:~$ ls -l
total 12
-r--r--r-- 1 berlin    ubuntu          0 May 31 14:46 devops.txt
-rw-r----- 1 ubuntu   ubuntu          12 May 31 14:49 notes.txt
drwxr-xr-x 2 ubuntu   ubuntu        4096 May 31 15:06 project
-rw-rw-r-- 1 professor heist-team      0 May 31 17:14 project-config.yml
-rw-rw-r-- 1 ubuntu   ubuntu          13 May 31 14:52 script.sh
-rw-rw-r-- 1 ubuntu   heist-team      0 May 31 17:09 team-notes.txt
ubuntu@ip-172-31-44-9:~$
```

Step 4: Create Directory

```
mkdir app-logs
```

Step 5: Change Owner and Group

```
sudo chown berlin:heist-team app-logs
```

Verify:

```
ls -ld app-logs
```

Note:

-d displays directory details.

```
ubuntu@ip-172-31-44-9:~$ mkdir app-logs
ubuntu@ip-172-31-44-9:~$ ls -l
total 16
drwxrwxr-x 2 ubuntu    ubuntu    4096 May 31 17:20 app-logs
-r--r--r-- 1 berlin    ubuntu    0 May 31 14:46 devops.txt
-rw-r----- 1 ubuntu    ubuntu    12 May 31 14:49 notes.txt
drwxr-xr-x 2 ubuntu    ubuntu    4096 May 31 15:06 project
-rw-rw-r-- 1 professor heist-team  0 May 31 17:14 project-config.yml
-rw-rw-r-- 1 professor heist-team  0 May 31 17:18 project-config1.yml
-rw-rw-r-- 1 ubuntu    ubuntu    13 May 31 14:52 script.sh
-rw-rw-r-- 1 ubuntu    heist-team  0 May 31 17:09 team-notes.txt
ubuntu@ip-172-31-44-9:~$ sudo chown berlin:heist-team app-logs
ubuntu@ip-172-31-44-9:~$ ls -l
total 16
drwxrwxr-x 2 berlin    heist-team 4096 May 31 17:20 app-logs
-r--r--r-- 1 berlin    ubuntu    0 May 31 14:46 devops.txt
-rw-r----- 1 ubuntu    ubuntu    12 May 31 14:49 notes.txt
drwxr-xr-x 2 ubuntu    ubuntu    4096 May 31 15:06 project
-rw-rw-r-- 1 professor heist-team  0 May 31 17:14 project-config.yml
-rw-rw-r-- 1 professor heist-team  0 May 31 17:18 project-config1.yml
-rw-rw-r-- 1 ubuntu    ubuntu    13 May 31 14:52 script.sh
-rw-rw-r-- 1 ubuntu    heist-team  0 May 31 17:09 team-notes.txt
ubuntu@ip-172-31-44-9:~$
```

Task 5: Recursive Ownership

Step 1: Create Directory Structure

```
mkdir -p heist-project/vault
```

```
mkdir -p heist-project/plans
```

Step 2: Create Files

```
touch heist-project/vault/gold.txt
```

```
touch heist-project/plans/strategy.conf
```

Step 3: Create Group

```
sudo groupadd planners
```

Step 4: Recursive Ownership Change

```
sudo chown -R professor:planners heist-project/
```

```
root@ip-172-31-44-9:/home/ubuntu# ls -l
total 20
drwxrwxr-x 2 berlin    heist-team 4096 May 31 17:20 app-logs
-r--r--r-- 1 berlin    ubuntu      0 May 31 14:46 devops.txt
drwxrwxr-x 4 ubuntu    ubuntu     4096 May 31 17:30 heist-project
-rw-r----- 1 ubuntu    ubuntu      12 May 31 14:49 notes.txt
drwxr-xr-x 2 ubuntu    ubuntu     4096 May 31 15:06 project
-rw-rw-r-- 1 professor heist-team   0 May 31 17:14 project-config.yml
-rw-rw-r-- 1 professor heist-team   0 May 31 17:18 project-config1.yml
-rw-rw-r-- 1 ubuntu    ubuntu      13 May 31 14:52 script.sh
-rw-rw-r-- 1 ubuntu    heist-team   0 May 31 17:09 team-notes.txt
root@ip-172-31-44-9:/home/ubuntu# su ubuntu
ubuntu@ip-172-31-44-9:~$ sudo chown -R professor:planners heist-project/
ubuntu@ip-172-31-44-9:~$ ls -l
total 20
drwxrwxr-x 2 berlin    heist-team 4096 May 31 17:20 app-logs
-r--r--r-- 1 berlin    ubuntu      0 May 31 14:46 devops.txt
drwxrwxr-x 4 professor planners  4096 May 31 17:30 heist-project
-rw-r----- 1 ubuntu    ubuntu      12 May 31 14:49 notes.txt
drwxr-xr-x 2 ubuntu    ubuntu     4096 May 31 15:06 project
-rw-rw-r-- 1 professor heist-team   0 May 31 17:14 project-config.yml
-rw-rw-r-- 1 professor heist-team   0 May 31 17:18 project-config1.yml
-rw-rw-r-- 1 ubuntu    ubuntu      13 May 31 14:52 script.sh
-rw-rw-r-- 1 ubuntu    heist-team   0 May 31 17:09 team-notes.txt
ubuntu@ip-172-31-44-9:~$
```

Meaning:

-R = Recursive

Changes ownership of the directory and everything inside it.

Step 5: Verify

```
ls -lR heist-project/
```



```
ubuntu@ip-172-31-44-9:~$ ls -lR heist-project/  
heist-project/  
total 8  
drwxrwxr-x 2 professor planners 4096 May 31 17:30 plans  
drwxrwxr-x 2 professor planners 4096 May 31 17:30 vault  
  
heist-project/plans:  
total 0  
-rw-rw-r-- 1 professor planners 0 May 31 17:30 strategy.conf  
  
heist-project/vault:  
total 0  
-rw-rw-r-- 1 professor planners 0 May 31 17:30 gold.txt  
ubuntu@ip-172-31-44-9:~$
```

heist-project/plans:

```
-rw-r--r-- 1 professor planners 0 May 27 strategy.conf
```

Task 6: Practice Challenge

Step 1: Create Users

```
sudo useradd tokyo
```

```
sudo useradd berlin
```

```
sudo useradd nairobi
```

Note:

If users already exist, ignore the warning.

Step 2: Create Groups

```
sudo groupadd vault-team
```

```
sudo groupadd tech-team
```

Step 3: Create Directory

```
mkdir bank-heist
```

Step 4: Create Files

```
touch bank-heist/access-codes.txt
```

```
touch bank-heist/blueprints.pdf
```

```
touch bank-heist/escape-plan.txt
```

Step 5: Set Ownership

For access-codes.txt

```
sudo chown tokyo:vault-team bank-heist/access-codes.txt
```

For blueprints.pdf

```
sudo chown berlin:tech-team bank-heist/blueprints.pdf
```

For escape-plan.txt

```
sudo chown nairobi:vault-team bank-heist/escape-plan.txt
```

Step 6: Verify

```
ls -l bank-heist/
```

Expected Output:

```
-rw-r--r-- 1 tokyo vault-team access-codes.txt
```

```
-rw-r--r-- 1 berlin tech-team blueprints.pdf
```

```
-rw-r--r-- 1 nairobi vault-team escape-plan.txt
```

```
ubuntu@ip-172-31-44-9:~/bank-heist$ ls -l
total 0
-rw-rw-r-- 1 tokyo  vault-team 0 May 31 17:48 access-codes.txt
-rw-rw-r-- 1 berlin tech-team 0 May 31 17:48 blueprints.pdf
-rw-rw-r-- 1 nairobi tech-team 0 May 31 17:48 escape-plan.txt
ubuntu@ip-172-31-44-9:~/bank-heist$
```

Commands Used

- `ls -l`
- `sudo chown`
- `sudo chgrp`
- `sudo groupadd`
- `sudo useradd`
- `touch`
- `mkdir`

What I Learned

- Owner controls the file.
 - Groups allow shared access.
 - `chown` changes ownership.
 - `chgrp` changes group ownership.
 - `-R` changes ownership recursively.
-

Important DevOps Real Use Cases

Scenario	Why Ownership Matters
Jenkins builds	Jenkins user must own workspace
Nginx logs	Web server needs correct permissions
Docker volumes	Container access issues occur if ownership is wrong
Shared team folders	Group access management
CI/CD artifacts	Deployment user ownership

Most Important Commands to Remember

```
ls -l
```

```
sudo chown user file
```

```
sudo chgrp group file
```

```
sudo chown user:group file
```

```
sudo chown -R user:group directory/
```

Quick Revision Trick

Command	Meaning
chown	Change owner
chgrp	Change group
-R	Recursive
ls -l	Show ownership and permissions